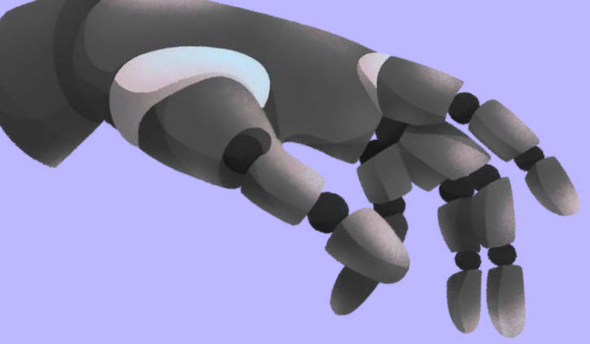
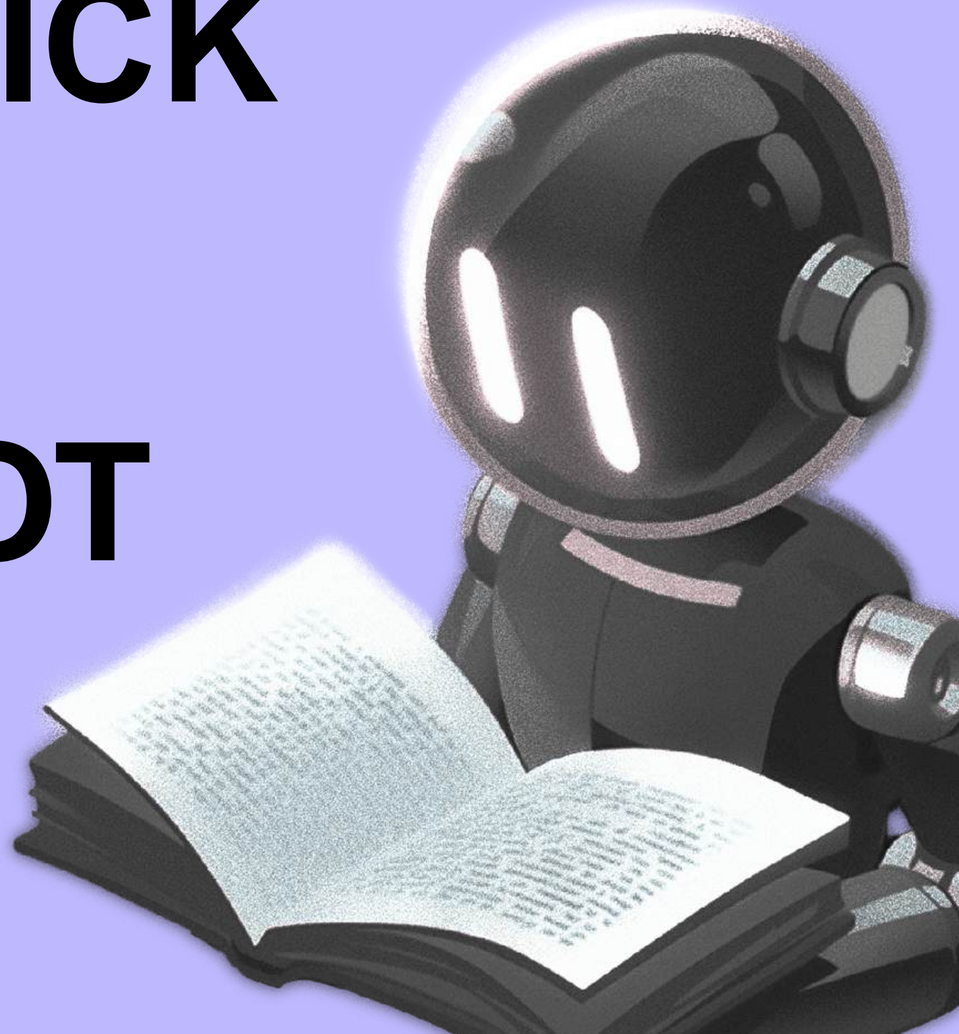


COLOR - BASED PICK & DELIVERY ROBOT



GROUP 2

Nethsara Anjula

Mishel Bashini

Ravindu Udara

Thiranjaya Chamudith

Supervisor - MR. Jude Fernando

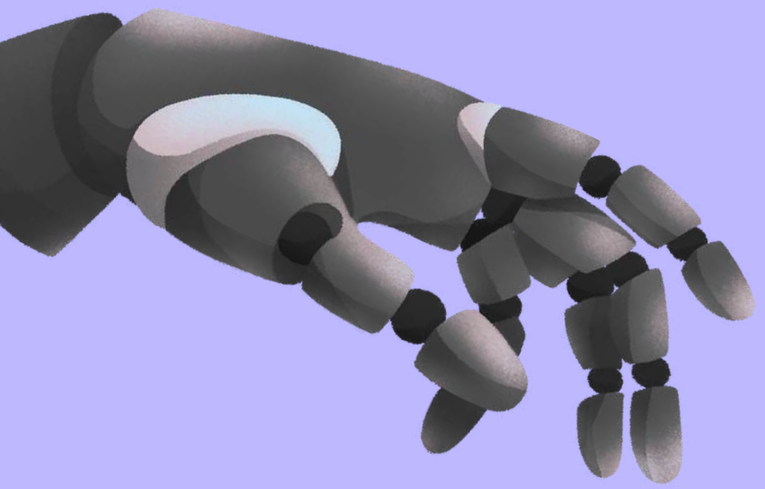
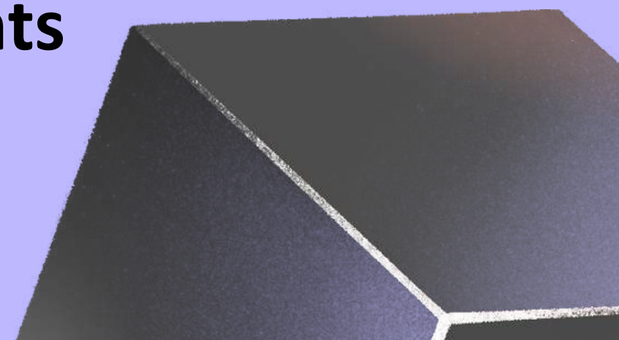
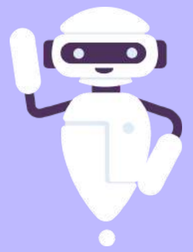


TABLE OF CONTENT

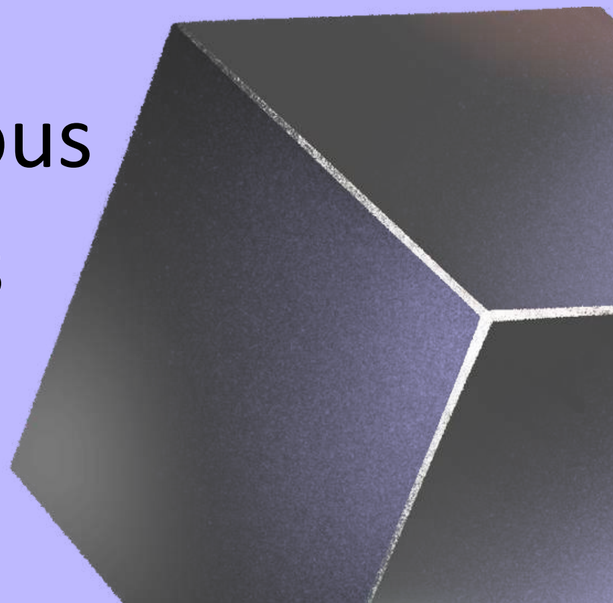
- Introduction
- Problem Statement
- Project Objectives
- Project Scope
- System Block Diagram
- Hardware Components
- Working Principle
- Design & Fabrication
- Testing & Results
- Application
- Advantages
- Limitation
- Future Improvements
- Conclusion



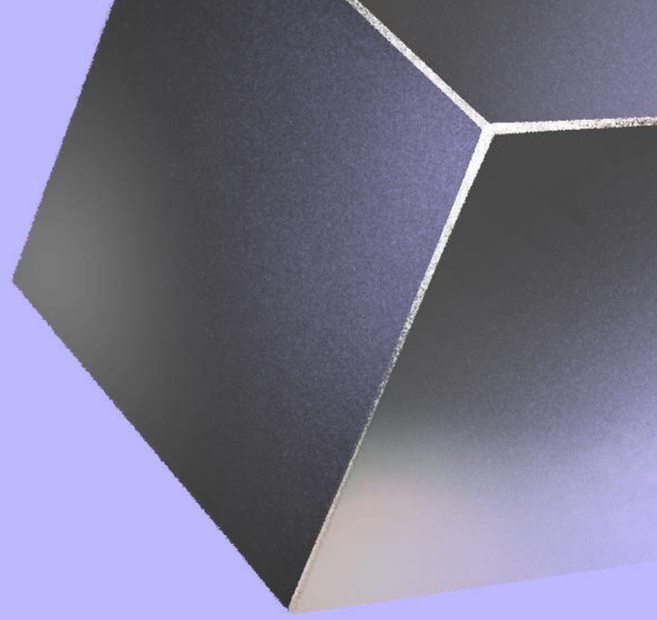
INTRODUCTION



- Automation plays a major role in modern industries
- AGVs are used to reduce human effort and increase accuracy
- This project focuses on an autonomous vehicle capable of pick-and-place operations

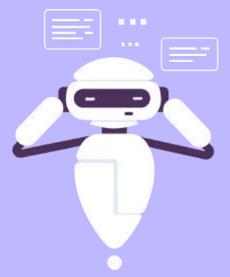


PROBLEM STATEMENT

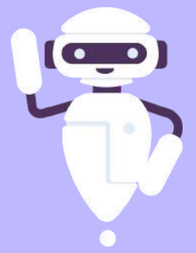


- Manual material handling is time-consuming and unsafe
- Human error reduces efficiency
- A low-cost automated solution is needed for material transportation and handling.

PROJECT OBJECTIVES



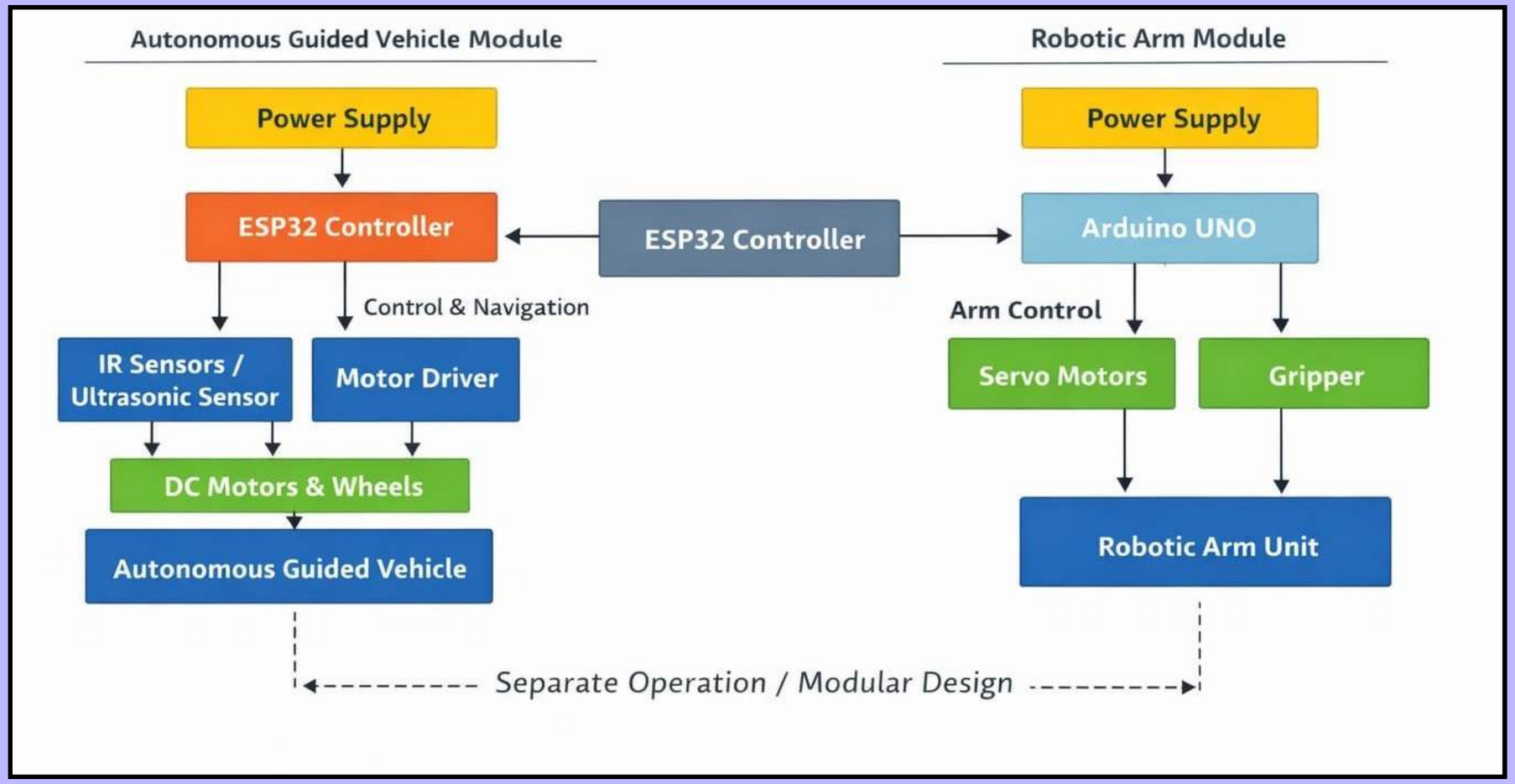
- Design and build a hand-operated vehicle
- Integrate a robotic arm for picking and placing tasks
- Use sensors for navigation and obstacle detection
- Improve efficiency and safety



PROJECT SCOPE

- Line following
- Obstacle detection using IR / ultrasonic sensors
- Robotic arm movement control
- Microcontroller-based system (ESP32)
- Indoor operation

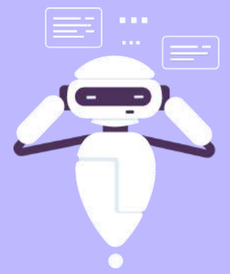
SYSTEM BLOCK DIAGRAM



HARDWARE COMPONENTS

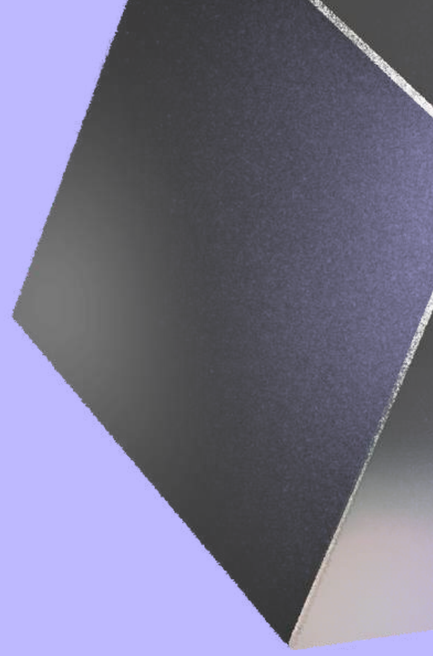
- ESP32 Microcontroller
- IR Sensors
- Ultrasonic Sensor
- Motor Driver Module
- DC Motors & Wheels
- Servo Motors (Robot Arm)
- Battery & Chassis

WORKING PRINCIPLE

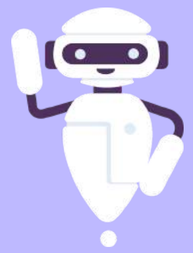


- Sensors detect the path and obstacles
- ESP32 processes sensor data
- Motor driver controls wheel movement
- Robotic arm automatically picks and places objects
- Arduino uno send PWM signals to server motors to control arm movements

DESIGN & FABRICATION



- Chassis designed for stability
- 3D printing used for chassis & arm parts
- Proper mounting of motors and sensors

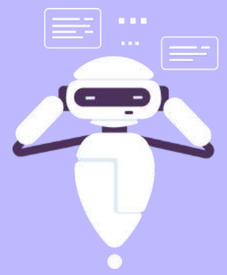


TESTING & RESULTS

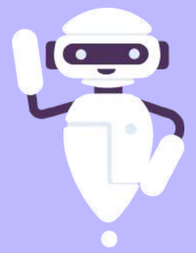
- The following line was successfully tested
- Obstacle detection is working correctly
- Selection and location implementation tested
- Minor alignment and calibration issues resolved

APPLICATION

- Warehouses
- Manufacturing industries
- Hospitals
- Smart factories
- Educational purposes



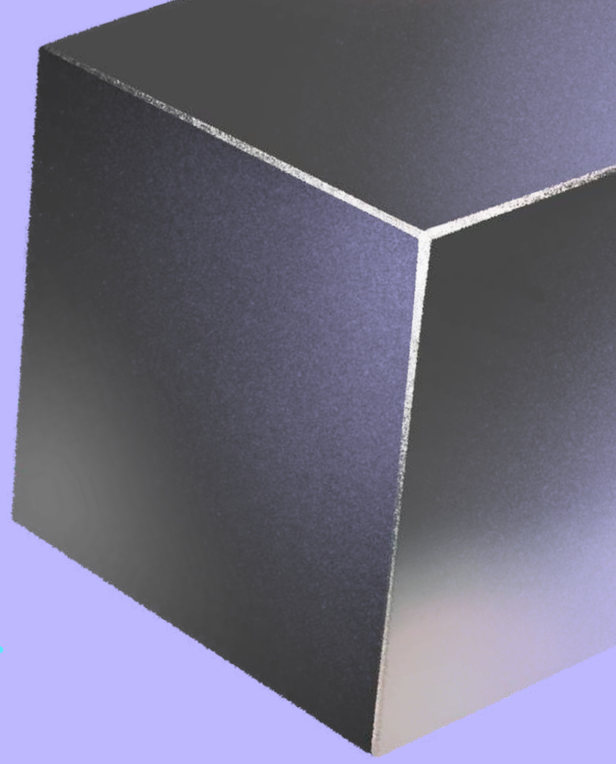
ADVANTAGES



- Reduces human effort
- Improves accuracy
- Low-cost solution
- Safe and efficient

LIMITATION

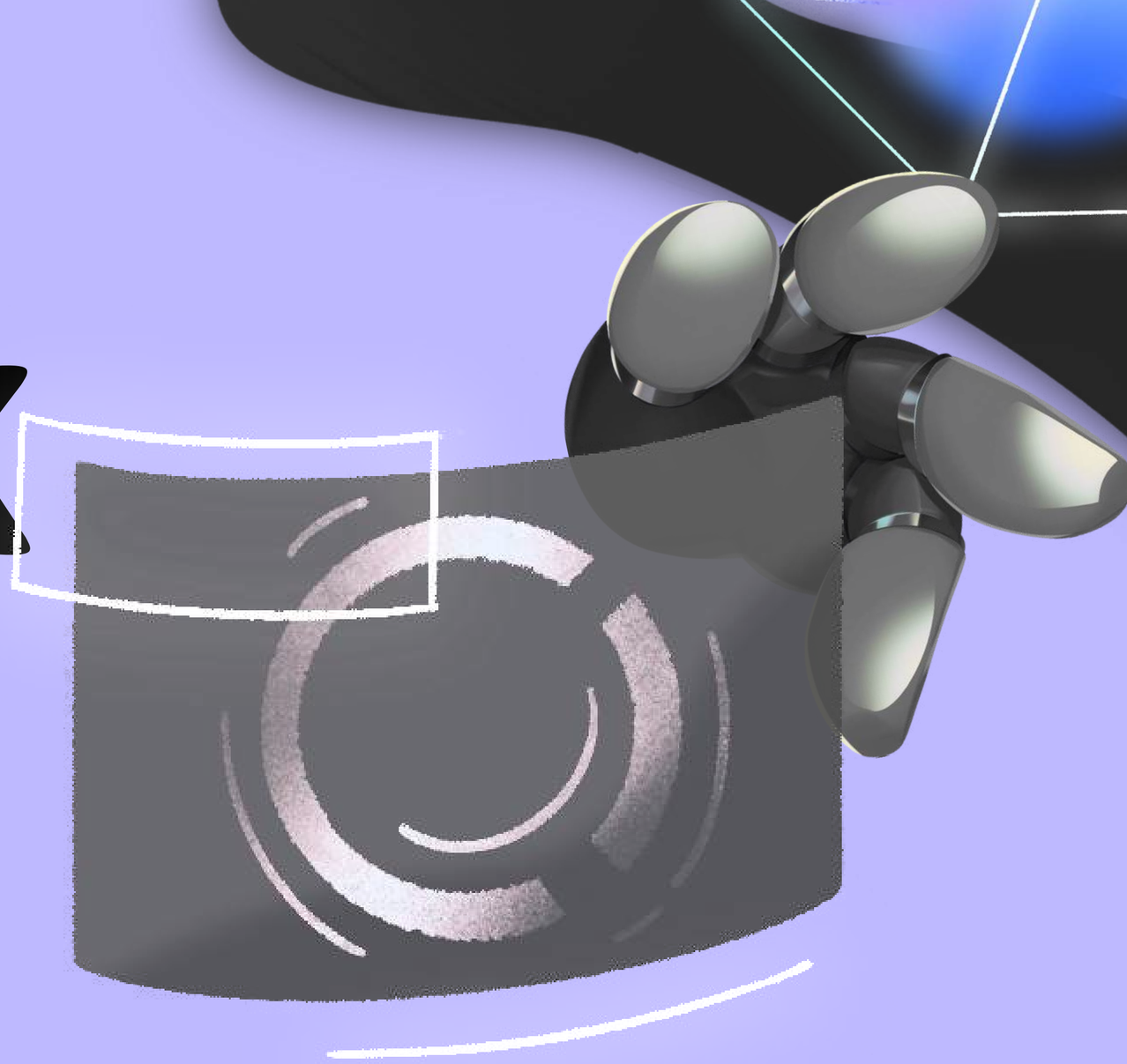
- Limited load capacity
- Indoor use only
- Requires calibration
- Battery life constraints



FUTURE IMPROVEMENTS

- AI-based navigation
- Camera integration
- Wireless monitoring
- Higher payload capacity
- IoT integration

CONCLUSION

- An AGV with a robotic arm was successfully designed and tested
- The project meets its objectives
- Demonstrates the practical application of automation and embedded systems

THANK YOU